

Airbnb Pricing Dashboard — Data ERD & Semantic Model Baseline

Prepared from the authoritative Updated Detailed Task Breakdown workbook. This document is the baseline for the model-building phase; it does not invent physical source columns that are not specified in the workbook.

1. Intended Star Schema

Entity	Role	Purpose / Grain	Key	Cardinality
Dim_Host	Dimension	Host-level descriptive attributes	host_id	1
Dim_Listing	Dimension	Listing/property attributes; linked to host	listing_id	1
Dim_Date	Dimension	Reporting calendar/date attributes	date	1
Fact_Calendar	Fact	Calendar availability observations at calendar-date grain	listing_id + date (with calendar_id if applicable)	*
Fact_Review	Fact	Review observations at review grain	review_id (with listing_id/date as foreign keys where applicable)	

2. Relationship Map

From	To	Cardinality	Relationship Key	Filter Intent
Dim_Host	Dim_Listing	1 : *	host_id	Host dimension filters listings
Dim_Listing	Fact_Calendar	1 : *	listing_id	Listing filters calendar observations
Dim_Listing	Fact_Review	1 : *	listing_id	Listing filters review observations
Dim_Date	Fact_Calendar	1 : *	date	Date filters calendar observations
Dim_Date	Fact_Review	1 : *	date	Date filters review observations, where review date is modeled

Logical ERD:

Dim_Host (1) → Dim_Listing (*) → Fact_Calendar (*)

Dim_Listing (1) → Fact_Review (*)

Dim_Date (1) → Fact_Calendar (*) and, where the review date is modeled, → Fact_Review (*)

3. Modeling Rules / Validation Baseline

Rule	Requirement
Architecture	Use a star-schema-oriented semantic model with dimensions filtering facts.
Cardinality	Relationships must be validated against actual key profiling before being accepted.
Filter direction	Use single-direction dimension-to-fact filtering wherever possible; avoid ambiguous filter paths.
Grain	Confirm the grain of every source/model table before creating relationships.
Keys	Validate uniqueness on dimension keys and referential integrity from fact foreign keys.
Date table	Create and validate Dim_Date; configure chronological sorting and use it consistently for reporting.
Technical fields	Hide technical/surrogate keys from report users where appropriate.
Display organisation	Use display folders for production measures and keep the field pane business-friendly.
Reconciliation	Reconcile Source → Raw → Clean → Model → DAX → Visual before the relevant gate can pass.

4. Important Scope Controls

Item	Rule
Unavailable calendar days	Must not be treated as confirmed bookings/rented days because unavailable can include host-blocked inventory.
Revenue	Confirmed/realised revenue and unsupported projected revenue are outside the current scope.
Occupancy	Confirmed occupancy is outside the current scope.
What-If scenarios	Occupancy/seasonal revenue scenario modelling is not part of the approved implementation.
Causal claims	Observational analysis must not be presented as causal evidence.

5. Model Gate — Required Evidence Before Dashboard Work

The model phase is not considered complete merely because the diagram exists. Before moving to DAX/report construction, the following must be evidenced and reviewed: table grains; key uniqueness; cardinality; relationship direction; orphan-record checks; Dim_Date coverage; field organisation; and basic model-performance checks. The project control rule is PASS/APPROVED before the next phase is permitted.

Source note: This ERD reflects the model entities and modelling controls defined in the Updated Detailed Task Breakdown. Exact source column inventories and physical data types should be confirmed during the data-understanding/quality phase rather than guessed here.